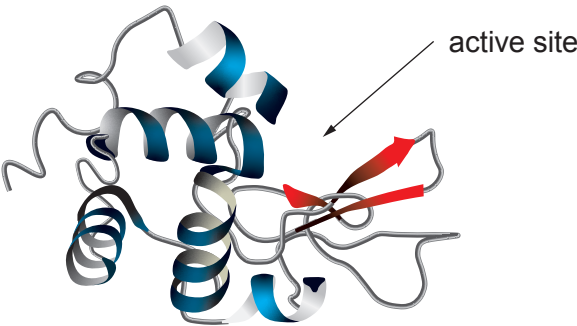


Answer **all** questions.

1. Lysozyme is an enzyme found in saliva and tears. It hydrolyses the carbohydrates in bacterial cell walls. Lysozyme, shown in **Image 1.1**, is a single polypeptide containing 129 amino acids.

Image 1.1 Ribbon diagram of lysozyme



- (a) Lysozyme shows primary, secondary and tertiary structure.

(i) State what is meant by primary structure.

[1]

.....

.....

- (ii) Use the information given and your own knowledge to **complete Table 1.2** to explain how the secondary and tertiary structures of lysozyme are illustrated in **Image 1.1**.

[2]

Table 1.2

Level of structure	Explanation
Secondary	
Tertiary	



Examiner
only

(iii) The cytoplasm of bacterial cells is **hypertonic** to saliva and tears. Suggest how the destruction of the cell wall of bacteria by lysozyme results in the death of the bacteria. [3]

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(iv) Lysozyme works by an induced fit mechanism. Explain what this means. [1]

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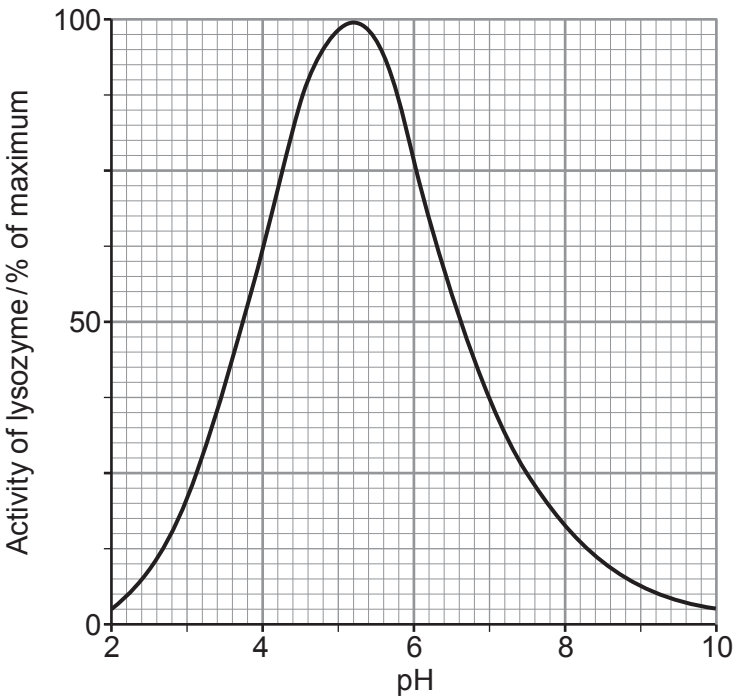
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(b) The effect of pH on the activity of lysozyme is shown in **Graph 1.3**.

Graph 1.3



(i) Describe and explain the effect of pH on the activity of lysozyme. [3]

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(ii) Lysozyme has a wider range of pH activity than many other enzymes.

I. Use **Graph 1.3** to find the range of pH over which the activity of lysozyme is more than 50% of the maximum. [1]

Range = to

II. Use the information in the question to suggest **one** advantage of lysozyme having a wider range of pH activity than many other enzymes. [1]

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